

Your review report

Manuscript

Geo-ViT: A Geometry-Aware Vision Transformer Framework for Underwater Image Enhancement

Feedback for the author(s)

Review file(s)

No files added.

Comments to the author(s)

The authors have presented in the manuscript an underwater image enhancement framework that can combine geometry-aware feature extraction, gated local-context modeling, transformer-based global attention, and residual learning. The proposed components and framework are aligned with the stated objective of improving the colour correction while preserving the local structural details. The benchmarking, qualitative comparisons, ablation studies, and efficiency analysis together give a broad evaluation. The earlier editorial request has been duly addressed. The direct links have been added to the Data Availability statement. Some technical clarifications would be appreciated –

1. Equation 2 adds the geometry map into the encoder feature directly. What if their channel dimensions are different, how will that affect the calculations? On the other hand, equation 3 separately defines a projected geometry feature. My understanding is that the formulation should consistently use the projected feature.
2. Good to have in equation 4 – define the decoder input and encoder stage more clearly.
3. The Charbonnier loss can be written unambiguously with all the parameters and perceptual loss details specified.
4. There is not enough clarity on whether the Sobel filters are fixed, initialized from Sobel kernels, or fully learnable.

Geo-ViT achieves strong PSNR results, but is not outperforming every method on every metric. So claims like "outperforms all models," "best-in-class," etc. should be moderated. The training protocol, number of runs, random seeds, baseline reproduction procedure, and runtime measurement conditions should also be clarified. In the future, the authors can explore lightweight attention, video consistency, unseen-water domain adaptation, downstream

detection or segmentation tasks.

Recommendation: Minor revision, subject to equations' correction or clarification, clarification of the experimental protocol, moderation of claims, and professional language editing.

Confidential feedback for the Editor

Your recommendation

- *Revise*

Is the study design appropriate to answer the research question (including the use of appropriate controls), and are the conclusions supported by the evidence presented?

- *No, but these points can be addressed with revisions*

Comments

The overall design is appropriate, but some conclusions are overstated and the experimental protocol requires clarification.

Are the methods sufficiently described to allow the study to be repeated?

- *No, but these points can be addressed with revisions*

Comments

Important details remain unclear, including the exact training strategy across datasets, random seeds, number of runs, baseline reproduction procedure, perceptual-loss configuration, and runtime measurement conditions.

Is the use of statistics and treatment of uncertainties appropriate?

- *No*

Comments

The manuscript reports single performance values without repeated-run statistics, variability, confidence intervals, or uncertainty estimates. This is especially relevant where differences between methods are small.

Has guidance been provided on how overstated claims should be rewritten?

- *I have highlighted which claims are inaccurate*

Comments

Some replacements can be like this – Geo-ViT achieves strong reconstruction accuracy across the evaluated datasets while maintaining competitive perceptual quality.

Is the presentation of the work clear?

- *No, it needs some language corrections before being published*

Comments

The work is understandable, but grammar, terminology, equation descriptions, typographical errors, and promotional wording need correction.

Are the images in this manuscript (including electrophoretic gels and blots) free from apparent manipulation?

- *Yes*

Confidential comments to the Editor